Experiment 7:

Use Hive to create, alter, and drop databases, tables, views, functions, and indexes.
Dataset:

```
1,Alice,20,F,85,CS
2,Bob,21,M,90,IT
3,Charlie,22,M,78,CS
4,Daisy,20,F,92,EE
```

1. Database:
Create a Database using student context

CREATE DATABASE student_db;

Use Database
USE student_db;

Alter Database
ALTER DATABASE student_db SET DBPROPERTIES ('description' = 'Database for student records');

Drop Database
Use `CASCADE` to delete all tables inside it.

2. Table Operations
Create Table

```
CREATE TABLE students (
   student_id INT,
   name STRING,
   age INT,
   gender STRING,
   marks INT,
   department STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';
```

Load Data into Table

LOAD DATA LOCAL INPATH '/path/to/student_data' INTO TABLE students;

Alter Table (add column)
ALTER TABLE students ADD COLUMNS (email STRING);

Queries on Table
SELECT * FROM students;

SELECT * FROM students WHERE marks > 85;

SELECT department, AVG(marks) AS avg_marks FROM students GROUP BY department;

SELECT SUBSTRING(name, 1, 3) FROM students;

Drop Table
DROP TABLE students;

3. View Operations
Create View
CREATE VIEW top_students AS
SELECT student_id, name, marks
FROM students
WHERE marks > 85;
Alter View (redefine)
ALTER VIEW top_students AS
SELECT student_id, name, marks, department
FROM students
WHERE marks > 85;

Queries on View
SELECT * FROM top_students;

SELECT department, COUNT(*) FROM top_students GROUP BY department;

Drop View
DROP VIEW top_students;

4. **Function Operations**
Hive UDFs (User Defined Functions) can be created using Java. For simplicity, we'll use built-in functions.

Use built-in function
SELECT UPPER(name) AS uppercase_name FROM students;

SELECT name, LENGTH(name) AS name_length FROM students;

Create Temporary Function (Java needed)
CREATE TEMPORARY FUNCTION to_upper AS 'com.hive.udf.ToUpperCase';

Drop Function
DROP TEMPORARY FUNCTION to_upper;

5. **Index Operations** (Used less commonly in modern Hive)
Create Index
CREATE INDEX idx_dept ON TABLE students (department)
AS 'COMPACT' WITH DEFERRED REBUILD;

Alter Index (Rename)
ALTER INDEX idx_dept ON students RENAME TO idx_department;

Drop Index
DROP INDEX idx_department ON students;

Export Table to File

```
INSERT OVERWRITE LOCAL DIRECTORY '/tmp/student_output'
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
SELECT * FROM students;
```